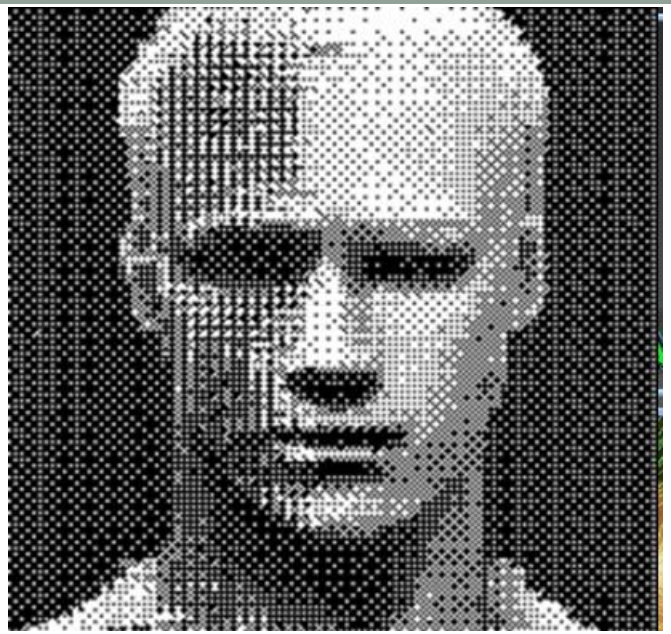


DIGITAL IMAGE PROCESSING

Dr. Manar Donia

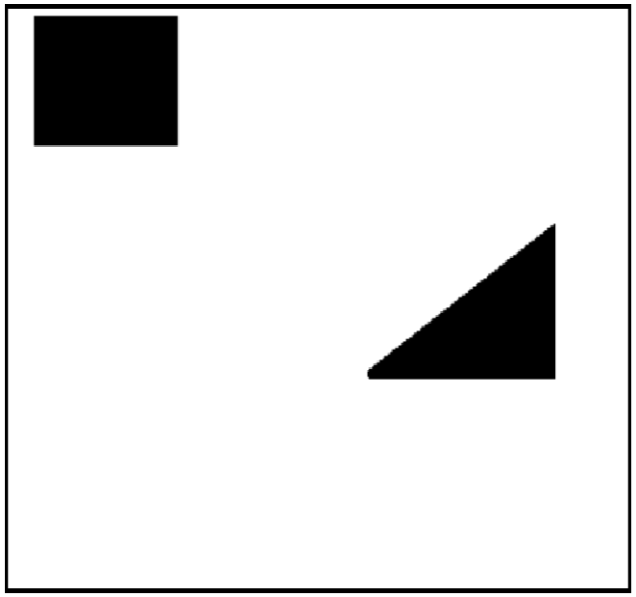
CMPn332 - CMPN332 & CMP 432

Digital Image representation

To convert Analog image into Digital we do:

Sampling: Digitizing the co-ordinate value.

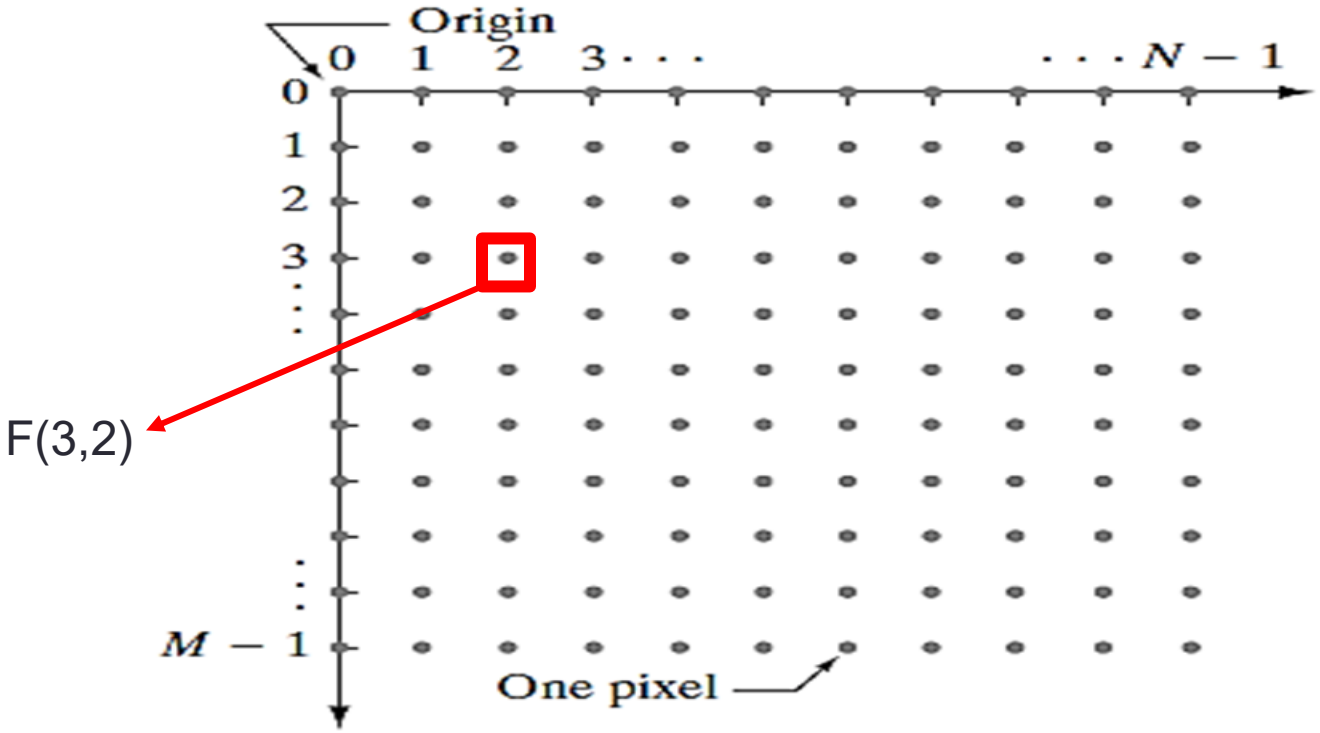
Quantization: Digitizing the amplitude value.



25 x 25 image matrix

```
Image Matrix =
[255, 210, 202, 205, 204, 201, 203, 206, 204, 219, 255, 255, 255, 255, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 253,  42,   2,   1,   0,   4,   1,   0,   1,  67, 252, 255, 255, 254, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  49,   1,   0,   3,   0,   0,   2,   2,  70, 255, 255, 249, 255, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  51,   0,   0,   0,   3,   0,   0,   3,  72, 255, 252, 255, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  47,   0,   6,   0,   0,   4,   0,   0,  75, 255, 254, 255, 253, 253, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  54,   0,   0,   2,   0,   3,   0,   0,  75, 249, 255, 255, 255, 255, 250, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 254,  40,   3,   1,   0,   3,   0,   0,   0,  65, 255, 253, 253, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255,  85,  39,  45,  39,  45,  43,  44,  43, 106, 252, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 253, 255, 255, 255, 254, 255, 139, 209;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 254, 255, 254, 255, 133,   0, 210;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 251, 255, 251, 255, 120,   0,   1, 213;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 112,   1,   1,   0, 215;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 253, 101,   0,   0,   0,   0, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 253, 255,  96,   0,   2,   3,   1,   0, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 254, 253,  83,   0,   2,   0,   0,   4,   0, 215;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 254, 253,  76,   0,   1,   0,   1,   0,   0,   2, 214;
 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 255, 255, 254, 227,  45,   0,   0,   0,   0,   0,   0,   0,   0, 211;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 255, 252, 206, 143, 155, 155, 155, 155, 155, 155, 155, 155, 240;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 252, 254, 254, 255, 251, 255, 255, 255, 255, 255, 255, 255, 255, 254;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 252, 255, 255, 255, 252, 252, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 252, 255, 253, 254, 255, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 251, 255, 255, 253, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 254;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 249, 254, 255, 252, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 254, 255, 255, 254, 254, 255, 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255, 255]
```

Digital Image representation



To refer to a specific pixel within the image matrix, we define its coordinate at row and column.

Digital Image representation

Image →8 x 8 pixels



F =

```
[255, 255, 252, 255, 255, 255, 255, 253;
 254, 255, 253, 254, 255, 253, 255, 255;
 255, 255, 255, 255, 255, 255, 253, 252;
 255, 250, 255, 254, 165, 145, 149, 204;
 251, 255, 254, 253,  26,   4,   0, 120;
 255, 255, 255, 254,  31,   0,   0, 120;
 252, 255, 252, 255,  87,  58,  56, 159;
 255, 253, 255, 255, 254, 255, 255, 254]
```

The amplitude values are defined by bits (pixel depth)

Pixel depth(k) (k=number of bits per pixel)	Maximum gray levels ($L = 2^k$)	gray level interval 0→(L-1)
1	2	0,1
2	4	0→3
3	8	0→7
8	256	0→255

Digital Image storage and transmission

- When considering image storage and transmission requirements, you must consider how much space an image occupies.

- For M x N digital image and k bits (pixel depth)
  - Number of bits required to store the image is

$$M*N*k$$

- For M x N digital image and L Maximum gray levels
  - Number of bits required to store the image is

$$M*N*\text{Log}_2(L)$$

Resolution

- Image resolution is the level of detail contained in an image. It is primarily determined by two distinct components: **spatial resolution** and **intensity resolution**.
  - Spatial resolution is a measure of the smallest discernible detail in an image.
    - Quantitatively, spatial resolution expressed as the number of pixels per unit distance, such as **dots per inch (dpi)**.
  - Intensity resolution refers to the smallest discernible change in intensity level.
    - Quantitatively, Intensity resolution is number of bits per pixel **(bpp)** needed to represent all possible intensity levels, usually is an integer power of two (2^k)

Spatial Resolution

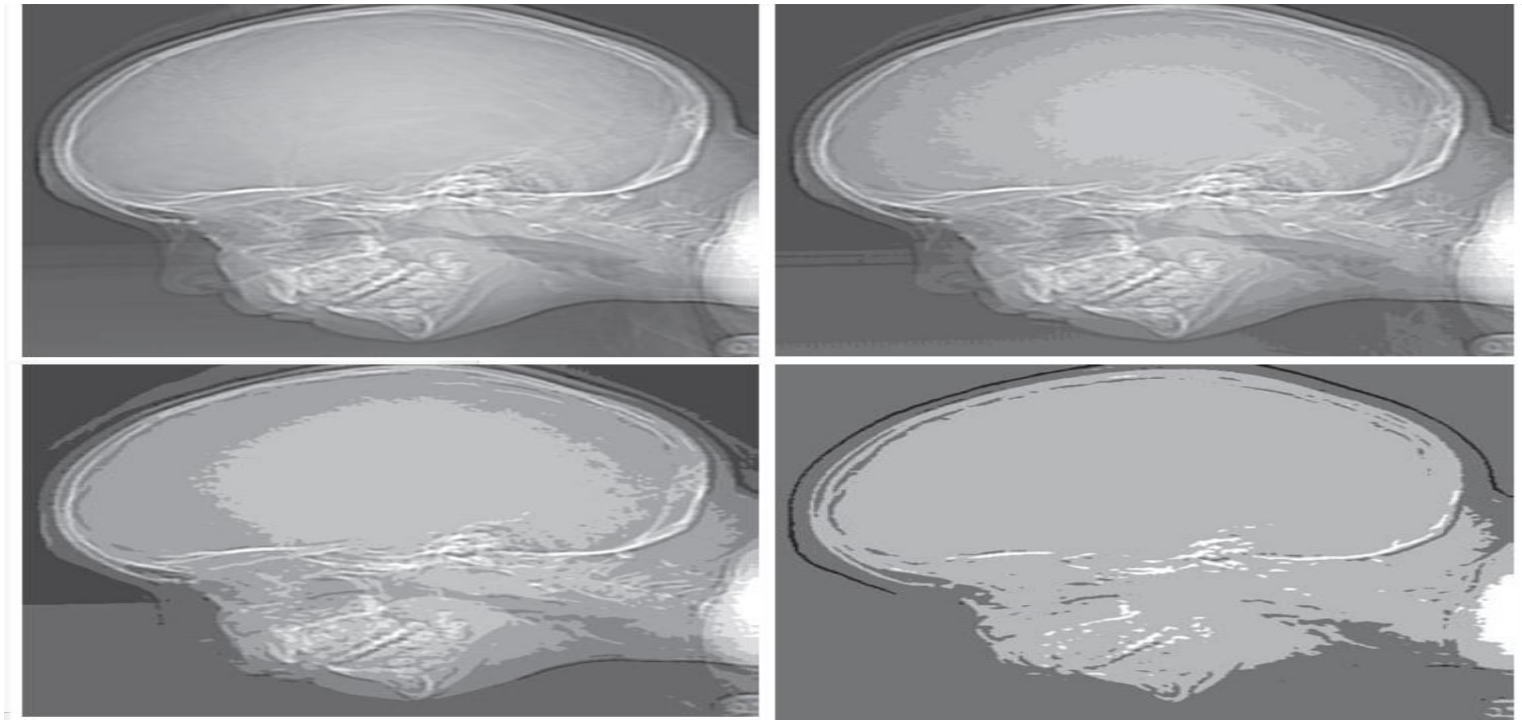
- Higher spatial resolution means more detail, sharpness, and clarity of image.



Effects of reducing spatial resolution.
The images shown are at: 930 dpi, 300 dpi, 150 dpi, and 72 dpi.

Intensity Resolution

- Higher intensity resolution means smoother gradients, better color accuracy, and more detailed tones.



Effects of reducing Intensity resolution. .
Image displayed in 256,16, 8, and 4 intensity levels, while keeping the spatial resolution Constant at 774 × 640.

Resolution







Spatial Resolution

- For high-resolution images, we must take into consideration.
  - Memory Requirements.
  - Processing time.
  - Transmission time.

Digital Image quality

Several factors influence the image quality, each contributing to how sharp, and detailed an image appears.

- Besides the image resolution, the **Dynamic range** is a key factor in determining the overall quality of an image.
  - **Dynamic range** describes the ratio between the maximum and minimum intensities in the Image (from the darkest to the brightest).



Example

For the given image matrix $F(x, y)$,determine the following parameters:

- Dynamic range
- Spatial resolution
- Intensity resolution

F(x,y)=

[190,	10,	107,	100;
120,	125,	120,	130;
115,	117,	112,	118;
130,	140,	145,	151]

- **Answer**
- Dynamic range = 10 : 190
- Spatial resolution = width × height in pixels.

  Width = 4, Height = 4. So **spatial resolution = 4 × 4 = 16 pixels**
- Intensity resolution = 8 bits

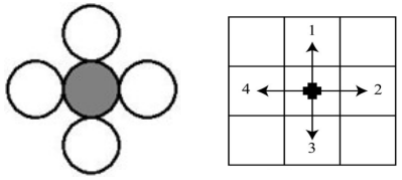
Relationships between pixels

- Pixel relationship helps define the structure, content, and the image's visual quality.
- They are vital in many tasks such as image enhancement, segmentation, edge detection, and feature extraction.
- key types of relationships between pixels:
  - Neighboring
  - Connectivity (adjacent)
  - Labelling (connected regions)
  - Distance measures

Neighboring

- N_4 neighbors (above, below, left and right)

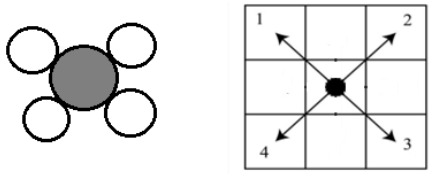
$$N_4(f(x,y)) = \{f(x-1,y), f(x+1,y), f(x,y-1), f(x,y+1)\}$$



4-Connected

- N_D neighbors (4 Diagonals)

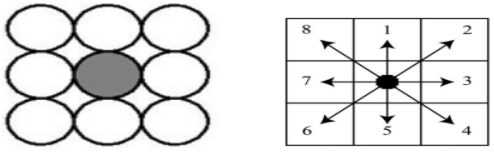
$$N_D(f(x,y)) = \{f(x-1,y-1), f(x+1,y-1), f(x-1,y+1), f(x+1,y+1)\}$$



D-Connected

N_8 neighbors (above, left, right, below, and 4 diagonals)

$$N8(f(x,y)) = \{N4 \cup ND\}$$



8-Connected

Neighboring

Image →8 x 8 pixels



F =

```
[255, 255, 252, 255, 255, 255, 255, 253;
 254, 255, 253, 254, 255, 253, 255, 255;
 255, 255, 255, 255, 255, 255, 253, 252;
 255, 250, 255, 254, 165, 145, 149, 204;
 251, 255, 254, 253,  26,   4,   0, 120;
 255, 255, 255, 254,  31,   0,   0, 120;
 252, 255, 252, 255,  87,  58,  56, 159;
 255, 253, 255, 255, 254, 255, 255, 254]
```

For the shown image matrix find N_4, N_D and N_8 neighbors for pixel F(5,4)

Answer

$$\underline{N_4(f(x,y)) = \{f(x-1,y), f(x+1,y), f(x,y-1), f(x,y+1)\}}$$

$$N_4(f(5,4)) = \{f(4,4), f(6,4), f(5,3), f(5,5)\} \rightarrow N_4(f(5,4)) = \{26,87,254,0\}$$

$$\underline{N_D(f(x,y)) = \{f(x-1,y-1), f(x+1,y-1), f(x-1,y+1), f(x+1,y+1)\}}$$

$$N_D(f(5,4)) = \{f(4,3), f(6,3), f(4,5), f(6,5)\} \rightarrow N_D(f(5,4)) = \{253,255,4,58\}$$

$$\underline{N_8(f(x,y)) = \{N_4 \cup N_D\}}$$

$$N_8(f(5,4)) = \{f(4,3), f(4,4), f(4,5), f(5,3), f(5,5), f(6,3), f(6,4), f(6,5)\}$$

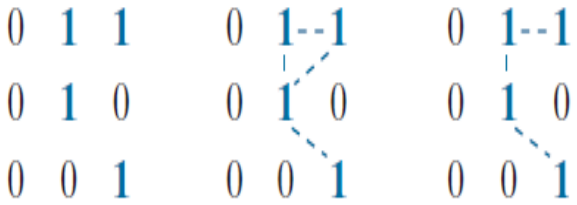
$$N_8(f(5,4)) = \{253, 26, 4, 0, 58, 87, 255, 254\}$$

Connectivity

- Two pixels are said to be connected(adjacent) if they are neighbors, and their gray levels satisfy a similarity (belong to the same connectivity set {V}).
- We consider three types of Connectivity (adjacency):
  - 4-*adjacency*. Two pixels *p* and *q* with values from *V* are 4-adjacent if *q* is in the set $N_4(p)$.
  - 8-*adjacency*. Two pixels *p* and *q* with values from *V* are 8-adjacent if *q* is in the set $N_8(p)$.
  - *m-adjacency* (also called *mixed adjacency*). Two pixels *p* and *q* with values from *V* are *m*-adjacent if
    - q is $N_4(p)$
    - or $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V

Connectivity

Mixed adjacency (m- adjacent) is a modification of 8-adjacency, and is introduced to eliminate the ambiguities that may result from using 8-adjacency.



a b c

FIGURE 2.28 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (c) *m*-adjacency.

Connectivity

- For the shown binary image matrix find the 4-*adjacency*, 8-*adjacency and* m-*adjacency* giving the connectivity set $V = \{1\}$ for pixel (0,0)

F(x,y)=

[0,	1,	0;
1,	1,	1;
0,	1,	0]

Answer

Value at (0,0) = **0**, which is **not** in $V = \{1\}$.

4-adjacency neighbors of (0,0) in V: ∅

8-adjacency neighbors of (0,0) in V: ∅

m-adjacency neighbors of (0,0) in V: ∅

All are empty sets because (0,0) is a background pixel for V={1}.

Connectivity

- For the shown binary image matrix find the 4-*adjacency*, 8-*adjacency and* m-*adjacency* giving the connectivity set $V = \{1\}$ for pixel (0,1)

F(x,y)=

[0,	1,	0;
1,	1,	1;
0,	1,	0]

Answer

Value at (0,1) = **1**, which is in $V = \{1\}$ so,

4-adjacency neighbors of (0,1) $\in V$: $\{f(1,1)\}$

8-adjacency neighbors of (0,1) $\in V$: $\{f(1,0),\ f(1,1), f(1,2)\}$

m-adjacency neighbors of (0,1) $\in V := \{f(1,1)\}$ is in N_4 (0,1) →automatically m-adjacent

: $N_D(0,1) \in V := \{f(1,0), f(1,2)\}$ ***check on connectivity***

$f(0,1) N4 \in V \rightarrow \{f(1,1)\}$
$f(1,0) N4 \in V \rightarrow \{f(1,1)\}$
$N_4(0,1) \cap N_4(1,1) \in V = \{f(1,1)=1\}$
because $f(1,1)\ in\ \{V\}$
$so\ it's\ not\ m\ adjacent$

$f(0,1) N4 \in V \rightarrow \{f(1,1)\}$
$f(1,2) N4 \in V \rightarrow \{f(1,1)\}$
$N_4(0,1) \cap N_4(1,2) = \{f(1,1)=1\}$ because
$f(1,1)\ in\ \{V\}$
$so\ it's\ not\ m\ adjacent$

Connectivity

For the shown image matrix find the find the 4-*adjacency*, 8-*adjacency and* m-*adjacency* giving the connectivity set connectivity $V = \{26,4,0,31,87,58,56\}$ for the pixel $f(5,4)$

```
[255, 255, 252, 255, 255, 255, 255, 253;
 254, 255, 253, 254, 255, 253, 255, 255;
 255, 255, 255, 255, 255, 255, 253, 252;
 255, 250, 255, 254, 165, 145, 149, 204;
 251, 255, 254, 253,  26,   4,   0, 120;
 255, 255, 255,   0,  31, 254,   0, 120;
 252, 255, 252, 255, 254,  58,  56, 159;
 255, 253, 255, 255, 254, 255, 255, 254]
```

Answer

Value at (5,4) = **31**, which $\in V = \{1\}$ so,

4-adjacency neighbors of (5, 4) $\in V := \{f(4,4), f(5,3)\} = \{26,0\}$

8−adjacency neighbors of (5, 4) $\in V := \{f(4,4), f(4,5), f(5,3), f(6,5)\}$

m-adjacency neighbors of (5,4) $\in V := \{f(4,4), f(5,3)\}$ is in N_4→automatically m-adjacent

: $N_D(5,4) \in V = \{f(4,5), f(6,5\}$ ***check on connectivity***

$f(5,4)N4 \in V \rightarrow \{f(4,4), f(5,3)\}$

$f(4,5)N4 \in V \rightarrow \{f(4,4), f(4,6), f(5,5)\}$

$N_4(5,4) \cap N_4(4,5) \in V = \{f(4,4) = 26\}$

$so\ it's\ not\ m\ adjacent$

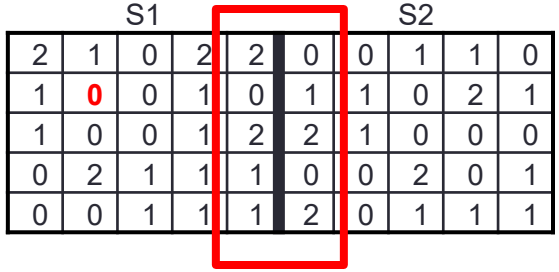
$f(5,4)N4 \in V \rightarrow \{f(4,4), f(6,4), f(5,5)\}$

$f(6,5)N4 \in V \rightarrow \{f(6,6)\}$

$N_4(5,4) \cap N_4(6,5) = \emptyset\ so\ it's\ m\ adjacent$

m-adjacency neighbors of (5,4) $\in V := \{f(4,4), f(5,3), f(6,5)\}$

•Consider the two image subsets, S1 and S2, shown in the following figure.
For connectivity set = {2}, determine whether these two subsets are
•4 adjacent
•8 adjacent
•m adjacent



Answer

Check on pixels S1(0,4), S1(2,4), S2(2,5) , S2(4,5) $\in V = \{2\}$ so,

4-adjacency neighbors

S1 (0,4) $\in V$: {S1(0,3)}

S1 (2,4) $\in V$: {S2(2,5)}

S2 (4,5) $\in V$: {ø}

Two regions, S1 and S2 are said to be *adjacent* if their union forms a connected set.
As, S1 (2,4) $\in V$: {S2(2,5)} so , S1 and S2 are 4 *adjacent*

8-adjacency neighbors

S1 and S2 are 4 *adjacent*, so 8-adjacency is automatically satisfied (since 4 ⊂ 8)

so , S1 and S2 are 8 *adjacent*

m-adjacency neighbors

Since S1 and S2 are 4 *adjacent*, so m-adjacency is automatically satisfied

so , S1 and S2 *m adjacent*

•Consider the two image subsets, S1 and S2, shown in the following figure. For connectivity set =$\{26,31,58,56,87,89\}$, determine whether these two subsets are

•4 adjacent
•8 adjacent
•m adjacent

S1				S2			
2	26	1	2	0	87	1	1
1	31	56	0	58	89	0	2
1	26	31	31	2	56	0	0
0	31	58	1	1	58	2	0
0	58	1	1	2	0	1	1

Answer

Check on pixels S1(2,3), S2(1,4) $\in V$ so,

4-adjacency neighbors

S1 (2,3) $\in V$: {S1(2,2)}

S2 (1,4) $\in V$: {S2 (1,5)}

As there is no union forms the connected set between S1 & S2 therefore they are *disjoint*.

8-adjacency neighbors

S1 (2,3) $\in V$:
{S1(2,2), S1(1,2), S1(3,2), S2(1,4)}

S1 and S2 are 8 *adjacent as* there is a union forms the connected set.

m-adjacency neighbors

As S1 and S2 are not 4 adjacent so check on diagonal

As S1(2,3), S2(1,4) are diagonal pixels and

S1 (2,3) $N4 \in V$: {S1(2,2)}

S2 (1,4) $N4 \in V$: {S2 (1,5)}

$N_4(2,3) \cap N_4(1,4) = \emptyset$
$so\ it's\ m\ adjacent$


Thank you!